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Advanced Machine Learning

Assignment 3: Time-Series Temperature Forecasting Using Deep Learning

A Comparative Analysis of GRU, LSTM, and Conv1D-GRU Models

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1. Introduction

Weather forecasting is an important application of time-series analysis because weather conditions change continuously and are influenced by previous observations. In this project, I used deep-learning models to predict temperature 24 hours ahead using historical weather measurements from the Jena Climate dataset.

The main goal was to compare different neural network architectures and identify which model produced the lowest forecasting error. I started with a simple persistence baseline and then experimented with GRU models, a stacked GRU, an LSTM, and a hybrid Conv1D-GRU model. Their performance was compared using Mean Absolute Error (MAE), where a lower value indicates more accurate predictions.

The results showed that the stacked GRU model achieved the lowest validation MAE of 2.318°C. It was therefore selected for final testing to determine whether its performance could generalize to unseen weather observations.

2. Dataset Description and Data Preparation

This project uses the Jena Climate dataset, which contains 420,551 weather observations recorded in Jena, Germany, at 10-minute intervals. The dataset includes 14 weather-related variables, such as temperature, atmospheric pressure, and humidity, along with date and time information. The initial examination showed no missing values.

To prepare the data for forecasting, I divided the observations chronologically into 50% training, 25% validation, and 25% testing. This approach helped prevent future observations from being used during model training. The weather variables were standardized using the mean and standard deviation calculated from the training data.

Each input sequence contained 120 hourly weather measurements, covering approximately five days of historical data. Since the original dataset recorded weather conditions every 10 minutes, I selected every sixth observation to obtain hourly measurements. Using these historical observations, the models predicted the temperature 24 hours after the final input measurement.

All models used the same data preparation process to maintain consistency and allow a fair comparison of their forecasting performance.

3. Deep Learning Models and Methods

To identify the most effective approach for temperature forecasting, I evaluated six forecasting methods, including a persistence baseline and five deep-learning models. Each model was assessed using validation Mean Absolute Error (MAE), which measures the average difference between predicted and actual temperatures in degrees Celsius. Lower MAE values indicate better forecasting performance.

Persistence Baseline: This simple method used the most recent temperature observation as the prediction for the future temperature. It provided a reference point for evaluating whether the neural networks offered meaningful improvements.

GRU (32 Units): I first implemented a Gated Recurrent Unit (GRU) network with 32 units to capture relationships between previous weather observations and future temperatures.

GRU (64 Units): I increased the number of GRU units from 32 to 64 to examine whether additional model capacity could improve prediction accuracy.

Stacked GRU (32 + 32 Units): This architecture used two GRU layers to learn more complex patterns from the weather sequences. Stacking recurrent layers allowed the network to process temporal information at multiple levels.

LSTM (32 Units): I also tested a Long Short-Term Memory (LSTM) network as an alternative recurrent architecture to compare its ability to capture longer-term dependencies.

Conv1D-GRU: Finally, I combined a one-dimensional convolutional layer with a GRU network. The convolutional component was designed to extract local patterns from the weather measurements before the GRU processed their temporal relationships.

All neural network models were evaluated using the same validation dataset. Early stopping was used to reduce unnecessary training and restore the best-performing model weights. The architecture with the lowest validation MAE was selected for final evaluation on the test dataset.

4. Results and Model Comparison

Forecasting Model	Validation MAE (°C)	Rank
Stacked GRU (32 + 32)	2.318	1
GRU (64 Units)	2.327	2
Conv1D-GRU	2.345	3
GRU (32 Units)	2.364	4
Persistence Baseline	2.443	5
LSTM (32 Units)	2.447	6

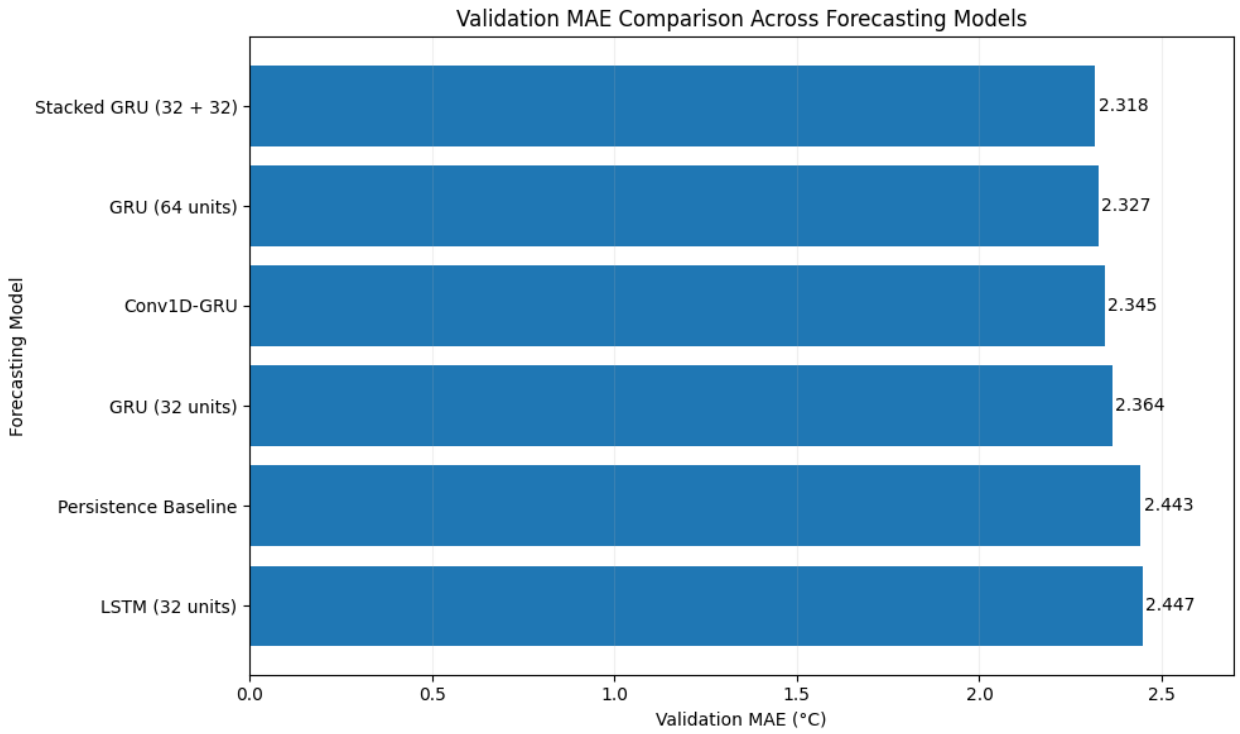


Figure 1. Comparison of validation MAE across six forecasting methods. Lower MAE values indicate better forecasting performance.

4.1 Interpretation of Validation Results

The validation results show that the stacked GRU model achieved the lowest MAE of 2.318°C, making it the best-performing model among the six methods tested. The GRU with 64 units followed closely with an MAE of 2.327°C, suggesting that increasing the network's capacity helped improve forecasting performance compared with the 32-unit GRU.

The Conv1D-GRU model also performed well, achieving a validation MAE of 2.345°C. This suggests that combining convolutional and recurrent layers can be useful for identifying patterns in weather data, although it did not outperform the stacked GRU in this experiment.

The LSTM model produced the highest validation MAE of 2.447°C, slightly exceeding the persistence baseline of 2.443°C. This indicates that using a more complex neural network does not automatically lead to better predictions.

Overall, four of the five deep-learning models outperformed the persistence baseline. The stacked GRU was selected for final testing because it achieved the lowest validation error, although its advantage over the 64-unit GRU was relatively small.

5. Final Test Performance

Table 2. Final Test MAE Comparison

Model	**Test MAE (°C)**
Persistence Baseline	2.620
Stacked GRU (32 + 32)	2.487

Lower test MAE indicates better forecasting accuracy on unseen observations.

5.1 Interpretation of Test Results

After comparing the validation results, I selected the stacked GRU model for final evaluation because it achieved the lowest validation MAE. On the test dataset, the stacked GRU recorded an MAE of 2.487°C, compared with 2.620°C for the persistence baseline.

This represents an improvement of approximately 5.1% in test MAE. In practical terms, the stacked GRU reduced the average absolute temperature prediction error by 0.133°C compared with simply using the most recent temperature as the forecast.

Although the test MAE was slightly higher than the validation MAE of 2.318°C, the stacked GRU still performed better than the baseline on unseen data. This suggests that the model learned useful patterns from historical weather observations and was able to apply them to new data.

These results support the selection of the stacked GRU as the best-performing model in this experiment. However, the improvement was relatively modest, showing that more complex models do not always produce large gains in forecasting accuracy.

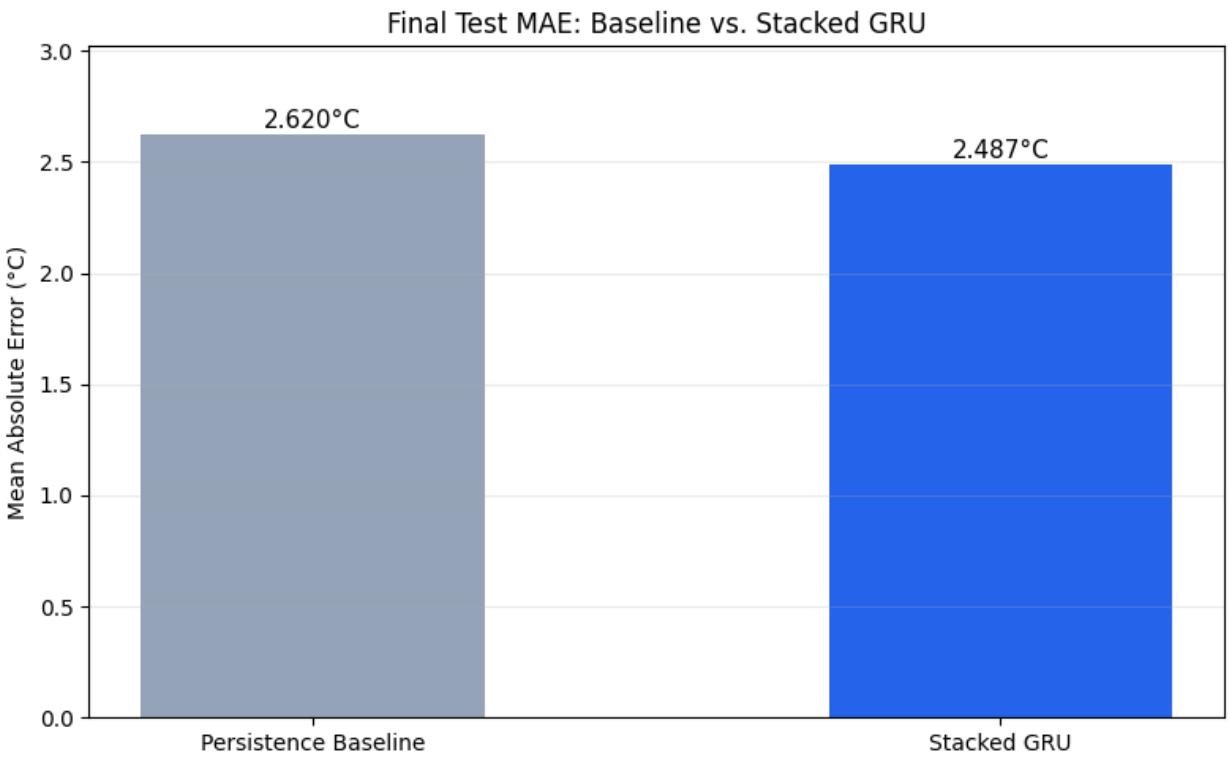


Figure 2. Final test MAE comparison between the persistence baseline (2.620°C) and stacked GRU model (2.487°C). The stacked GRU achieved approximately 5.1% lower prediction error on unseen data.

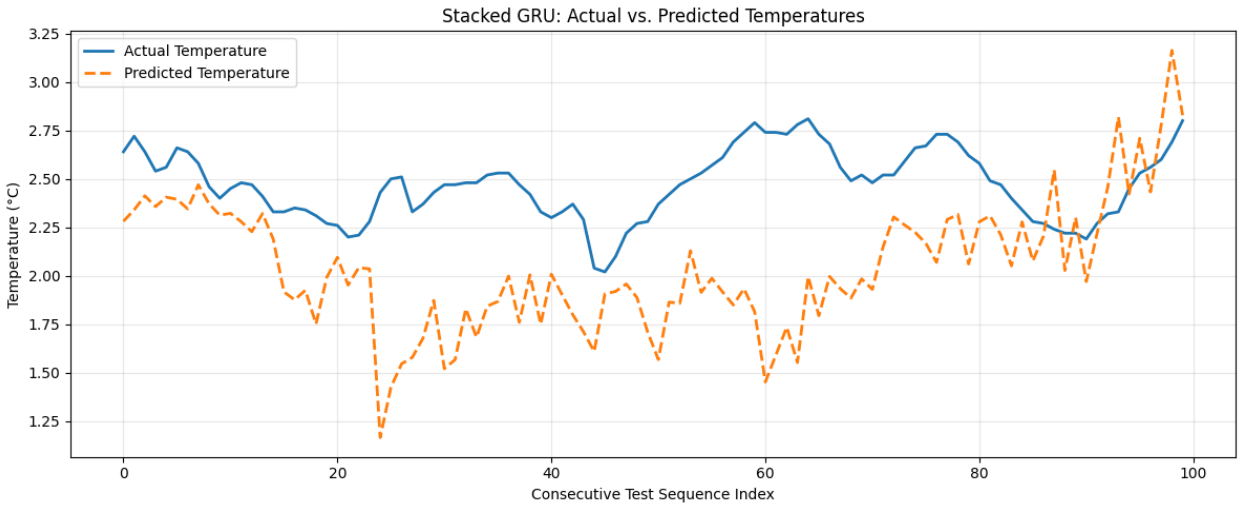


Figure 3. Actual vs. Predicted Temperature

This figure compares the actual temperature values with the predictions generated by the stacked GRU model on the test dataset. The model follows the general temperature patterns, although some differences remain between the predicted and observed values. These differences show that the model captures useful trends but does not predict every temperature change perfectly.

6. Discussion and Analysis

The results of this experiment show that model architecture plays an important role in time-series forecasting. Among the five deep-learning models tested, the stacked GRU achieved the lowest validation MAE of 2.318°C. Its performance suggests that using multiple recurrent layers can help capture useful patterns in historical weather observations.

Increasing the number of GRU units from 32 to 64 also improved validation performance, reducing MAE from 2.364°C to 2.327°C. However, the stacked GRU performed only slightly better than the 64-unit GRU, indicating that increasing model complexity does not always produce substantial improvements.

The Conv1D-GRU model achieved a validation MAE of 2.345°C, showing that combining convolutional and recurrent layers can provide competitive results. In contrast, the LSTM model recorded an MAE of 2.447°C, slightly worse than the persistence baseline. This suggests that the effectiveness of a particular architecture depends on the dataset and model configuration rather than its complexity alone.

An important observation was that the stacked GRU reached its best validation performance early in training, while later epochs did not improve the result. This highlights the value of early stopping in limiting unnecessary training and retaining the best-performing model weights.

On the test dataset, the stacked GRU reduced prediction error by approximately 5.1% compared with the persistence baseline. Although this improvement was modest, it demonstrates that the model was able to learn useful temporal relationships and outperform a simple forecasting approach on unseen data.

One limitation of this experiment is that the models were evaluated using a single chronological training, validation, and testing split. Different hyperparameters, training runs, or forecasting periods could produce different results. Future improvements could include systematic hyperparameter tuning, additional regularization techniques, and testing across multiple forecasting periods.

7. Conclusion

This assignment explored the use of deep-learning models for temperature forecasting using the Jena Climate dataset. Different GRU configurations, an LSTM model, and a hybrid Conv1D-GRU model were developed and compared with a persistence baseline to evaluate their forecasting performance.

Among the models tested, the stacked GRU with two 32-unit layers achieved the best validation MAE of 2.318°C. When evaluated on the test dataset, it achieved an MAE of 2.487°C, compared with 2.620°C for the persistence baseline. This represents a reduction of approximately 5.1% in test MAE compared with the persistence baseline.

Overall, the experiment demonstrated that recurrent neural networks can identify useful patterns in historical weather data and improve temperature predictions. However, the differences between some models were relatively small, showing that more complex architectures do not always guarantee better results.

This assignment also provided practical experience in preparing time-series data, training deep-learning models, comparing performance, and interpreting results.

References-

Max Planck Institute for Biogeochemistry. (n.d.). *Weather data from the Jena climate station*. https://www.bgc-jena.mpg.de/wetter/